

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVC504 - CONCRETE TECHNOLOGY

Time : 3 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the primary role of water in concrete?
- a) To increase workability
 - b) To activate hydration of cement
 - c) To reduce the cost of construction
 - d) To act as a binder

Bulking of sand is primarily caused by

- a) Temperature
- b) Silt content
- c) Moisture content
- d) Aggregate size

- 3 Which method is NOT used to measure the workability of concrete?
- a) Slump test
 - b) Vee Bee test
 - c) Vicat Apparatus test
 - d) Compaction factor test

Segregation in fresh concrete is defined as:

- a) Uniform distribution of water and aggregates
- b) Separation of coarse aggregates from the rest of the mix
- c) Increase in air content
- d) Decrease in workability

- 5 Which of the following is true about high-performance concrete?
- a) It has lower strength than conventional concrete
 - b) It is designed for better durability and workability
 - c) It is only used in low-strength applications
 - d) It requires no curing

- 6 Which admixture increases the workability of concrete without increasing water content?
- a) Retarder
 - b) Superplasticizer
 - c) Accelerator
 - d) Pozzolana

- 7 Green concrete refers to:
- a) High strength concrete
 - b) Concrete mixed with green pigment
 - c) Environmentally friendly concrete
 - d) Concrete that is not fully cured

- 8 Which of the following factors contributes to the corrosion of steel reinforcement in concrete?
- a) Sulphate attack
 - b) Chloride attack
 - c) Alkali-aggregate reaction
 - d) Thermal expansion

- 9 Characteristic strength of concrete is defined as strength below which not more than:
- a) 10 % of results fall
 - b) 5% of results fall
 - c) 50% of results fall
 - d) 25% of results fall

- W/C ratio mainly affects:
- a) Durability
 - b) Workability
 - c) Strength
 - d) All of the above

- 11 Creep in concrete is:
- a) Sudden deformation
 - b) Time-dependent deformation
 - c) Due to poor mix
 - d) Permanent expansion

- Non-destructive testing method used to determine concrete strength is:
- a) Core cutting
 - b) Rebound hammer test
 - c) Slump test
 - d) Compaction test

Q2. Solve the following.

- A) Explain the manufacturing process of Ordinary Portland Cement (OPC) with a neat flow diagram. 6
- B) Explain types of cement. 6

Q3. Solve the following.

- A) Define workability. Explain slum cone test for measuring workability 6

- B) Differentiate between volume batching and weight batching. 6
- Q4. Solve Any Two of the following.
- A) What are admixtures? Explain the classification and roles of different types of admixtures in concrete. 6
- B) Illustrate difference between accelerator and plasticizer. 6
- C) What is green concrete? Discuss its benefits and applications in sustainable construction. 6
- Q5. Solve Any Two of the following.
- A) What is Alkali Aggregate Reaction (AAR) and explain factors affecting on AAR. 6
- B) Define creep. Explain factors affecting Creep. 6
- C) Compare high-strength concrete and high-performance concrete. 6
- Q6. Solve Any Two of the following.
- A) Explain the process of concrete mix design. What are the factors influencing mix design? 6
- B) What are the commonly used non-destructive tests (NDT) for concrete? Describe any two in detail. 6
- C) Explain quality measurement in Concrete Construction. 6

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